Ecological Stewardship in Practice: Occupational Experience and Psychological Adaptation—An In-Depth Interview Study of a Forest Ranger at Tangjiahe National Nature Reserve

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Abstract

Tangjiahe National Nature Reserve is a key component of giant panda (*Ailuropoda melanoleuca*) habitat within the Minshan mountain range. Frontline forest rangers play an essential role in biodiversity monitoring, forest protection, wildlife management, and disaster early warning. This study draws on a semi-structured, in-depth interview with a ranger who has accumulated eleven years of service at Tangjiahe. Analyzing the interview across four dimensions—daily patrol practice, nature-based experience, professional development, and value identification—we examine the ranger's work content, psychological experience, and the meaning he derives from his profession.

The findings show that the ranger's work has gradually shifted from traditional tasks such as fire prevention and anti-poaching toward a more integrated portfolio: ecological monitoring, epizootic disease surveillance, observation of forest pests, and geological disaster early warning. His occupational role has correspondingly evolved from a singular "guardian" into a comprehensive ecological steward. Living in close contact with nature over the long run has cultivated a distinctive way of perceiving the natural world; recording wildlife imagery and conducting field observation have not only enriched his professional experience but also become important means of coping with solitude and regulating his psychological state. At the same time, mountain life brings professional loneliness, an information gap with the outside world, and the psychological pressure of having lived through natural disasters.

Further analysis indicates that the ranger's professional identification follows a developmental trajectory: from being driven by practical needs, to gradually developing an ecological interest, and finally arriving at identification with the value of his work. Family support has been a vital guarantee for his long-term commitment, and "doing what I love" has become his

central expression of what this role means to him. The individual experience of this Tangjiahe ranger offers an authentic portrait of frontline workers in protected areas and provides a concrete case for understanding occupational experience and value identification in ecological conservation work.

Keywords： Tangjiahe National Nature Reserve; forest ranger; ecological stewardship; occupational interview; professional identification

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1 Introduction

The report to the 20th National Congress of the Communist Party of China explicitly proposed that “Chinese-style modernization is the modernization of harmonious coexistence between human and nature,” and positioned “establishing a natural protected area system with national parks as the mainstay” as an important institutional arrangement for ecological civilization construction[1]. Against this macro backdrop, protected areas not only shoulder the vital tasks of ecological protection and biodiversity conservation but also carry functions such as the transmission of ecological culture[2]. The people who truly keep these protected areas running day by day are the frontline forest rangers who live for long stretches in the mountains, alongside wild plants and animals.

Tangjiahe National Nature Reserve, located in Qingchuan County, Sichuan Province, is an important habitat for the giant panda and other rare wild animals and sits within a global biodiversity conservation hotspot[3]. Since its establishment in 1978, the reserve has accumulated rich experience in species conservation and ecological management. However, with growing demand for biodiversity monitoring[3], the increasing frequency of extreme weather events[4], and the higher requirements that the national-park system places on frontline management, the content of a ranger’s work has gradually shifted from traditional patrol duties toward more comprehensive ecological protection[5][6]. At the same time, long-term life in the forest shapes the ranger’s occupational experience, psychological state, and value identification[7][8]

Existing research has primarily examined the operational mechanisms of protected areas and the effectiveness of ecological protection from the angles of protected-area administration, biodiversity conservation, and patrol-system construction[9]. Yet relatively little scholarly attention has been paid to the work experience, occupational adaptation, and psychological experience of the frontline forest rangers who carry out conservation work over the long run.

This study takes as its subject a frontline ranger at Tangjiahe with eleven years of patrol experience. Through a semi-structured, in-depth interview, the study proceeds along four dimensions—daily patrol, nature-based experience, professional development, and value identification—and attends to the specific tasks the ranger performs in daily work, the occupational challenges he faces, and how he understands the meaning of his work. By analyzing one

individual's experience, the paper hopes to provide a concrete case for understanding the occupational experience of frontline workers in protected areas.

2 Research Design and Methods

2.1 Research Site and Interviewee

This study selects Tangjiahe National Nature Reserve in Sichuan as the field site. The interviewee joined Tangjiahe Reserve in 2015 and has been working as a ranger ever since, accumulating eleven years of service. Before joining the reserve, he had spent many years on nuclear-power-station construction work in coastal provinces including Zhejiang, Fujian, and Shandong. He chose to return to his hometown and become a ranger because he wanted to balance work with family. The interviewee therefore brings two distinct professional backgrounds: one in engineering construction and one in ecological protection.

2.2 Data Collection and Analysis

This study uses a semi-structured, in-depth interview method to collect data. The interview guide was designed around four dimensions—daily patrol, stories of stewardship, professional development, and value identification—and covers topics such as occupational experience, patrol routines, psychological state, and disaster experience. The interview was conducted in the form of an online meeting, with the entire session recorded and then transcribed into text.

3 Overview of Tangjiahe National Nature Reserve and the Patrol System

3.1 Basic Information about the Reserve

Tangjiahe National Nature Reserve covers a total area of approximately 40,000 hectares. Its principal conservation targets include rare and endangered species such as the giant panda, the golden snub-nosed monkey (*Rhinopithecus roxellana*), the Sichuan takin (*Budorcas taxicolor tibetana*), and the Chinese monal (*Lophophorus lhuysii*), together with their habitats[10]. In 2014, Tangjiahe Nature Reserve was selected for the first batch of the IUCN Green List of Well-Managed Protected Areas, and is one of the places in the world with the highest rate of wildlife encounters within a low-elevation forest ecosystem[10].

3.2 Patrol System

According to the interviewee's account, Tangjiahe Reserve has now established a relatively mature patrol system. Protection stations must carry out routine patrol for no fewer than 15 days each month, supplemented by quarterly patrols that together produce periodic full coverage of the area. Field work strictly enforces the safety rule of "three or more persons traveling at once," prohibiting any solo operation.

4 Findings

4.1 Daily Patrol: From Traditional Custodial Work to Integrated Ecological Stewardship

4.1.1 Realistic Choice and Identity Adjustment in Occupational Transition

The interviewee had previously been engaged in nuclear-power-station construction and had long worked in coastal areas. When asked about the trigger for the career shift, he pointed to "being far from home and the inconvenience of visiting family" as the core practical factor behind considering a change, after which he chose to return to his hometown and join Tangjiahe Nature Reserve as a ranger.

Although the work environment changed, nuclear-power-station sites are also built in remote coastal areas, so the interviewee was not unfamiliar with working far from the city, which allowed him to adapt relatively quickly to the protected area's work environment.

4.1.2 The Diverse Tasks and Expertise Required by Patrol Work

In daily work, the ranger faces far more than mountain-patrol tasks. He also has to handle a wide range of ecological protection and safety-management assignments. The interviewee summarized the patrol duties as follows: observation and recording of wildlife traces and behavior; epizootic-disease surveillance and sampling analysis of dead animals; statistics on forest pests and diseases; inspection of geological-disaster hazard sites such as landslides and debris flows; and supervision of tourists who illegally cross restricted areas or wade in the water.

4.2 Everyday Stewardship: Nature Experience and Local Knowledge from Long-Term Coexistence

4.2.1 A Healing Bond: Therapeutic Experience and Professional Fulfillment from Being with Nature

Compared with the construction site, the patrol practice of living with nature over the long run has given the interviewee a distinctive professional experience. When comparing his two jobs, the interviewee noted that because daily patrols allow him to encounter different wild plants and animals, the reserve work is relatively more interesting. This kind of nature-based experience—wildlife observation, the changing seasons, the forest environment—gives him a sense of fulfillment he did not have in his previous job, and to a certain extent eases the fatigue of long-term field work.

4.2.2 Natural Imagery and Psychological Adjustment in an Off-Grid Environment

During deeper patrols such as the quarterly monitoring, rangers often have to enter "off-grid no-signal wilderness" for several days in a row, where there is no cellular signal. For someone accustomed to modern connected life, this is an enormous psychological test. The interviewee shared a vivid detail:

"When we run the quarterly monitoring, we might have to stay on the mountain for four or five days, completely in an area with no signal and no contact with the outside world. When you truly have no signal, sometimes you do feel it's very boring, time passes very slowly, and you get anxious. When we really have no internet and no signal, that's when you pull out your phone and look at the photos of plants and animals you shot during the day's work, and you think, well, this is still pretty interesting."

In the signal-less depths of the mountains, the wildlife and forest images he has recorded become a way for him to reconnect with nature and ease his feelings of loneliness.

4.2.3 Ecological Observation and Disaster Anticipation Based on Local Experience

Through long-term field practice, the ranger has gradually developed the ability to judge based on environmental observation and accumulated experience. For example, by observing unusual behavior in rivers and wildlife, he can develop an intuitive anticipation of extreme weather. He mentioned:

"Not long ago there was a particularly large King Ratsnake climbing up the road toward the mountain. After extremely heavy rain, the river would flood, and the burrow where the snake lived might be submerged, so it would head for higher slopes to migrate. From that we judged that heavy rain was coming. Sure enough, a very heavy downpour came shortly after, and the river rose."

This kind of keen intuition that links animal behavior to weather change is an important ability formed through long-term frontline observation and accumulated experience.

4.3 Occupational Adaptation: Loneliness, the Information Gap, and Risk Response

4.3.1 The Formation, Change, and Adaptation of Professional Loneliness

"Loneliness" was one of the more frequently mentioned words in the interview. The interviewee described the loneliness experienced during long-term patrol as a phased evolution: "from a sense of novelty, to tedium, to gradual adaptation, and on to embracing solitude." This trajectory suggests that professional loneliness is not a static psychological state but a process that is continually adapted to and reinterpreted over the course of a long career. Rangers must continually readjust their relationship with their work and with the environment over the long run.

4.3.2 The Information Gap and Social Connection in Mountain Life

The interviewee specifically mentioned that living in the mountains for a long time creates an "information gap" with the outside world: apart from communication with a fixed group of colleagues, his understanding of city life, popular culture, and social issues tends to lag behind. This "information gap" takes on a new form in the digital age: on the one hand, short videos and other new media provide rangers with a partial channel of "indirect contact with the outside world"; on the other hand, the absence of real face-to-face interaction means that information-level "access" cannot substitute for the psychological sense of belonging. This phenomenon shows that, although digital technology has lowered the threshold of information access, it cannot fully replace genuine human interaction.

4.3.3 Extreme Disaster Experience and the Psychological Impact on the Job

The sudden onset of geological disasters is the most direct threat to rangers' lives and safety. The interviewee described in detail a catastrophic debris-flow event he experienced in July 2024:

"In July 2024, it was already past 11 p.m. It had been raining hard for a day or two. The seven or eight of us at the station had already gone to bed. Suddenly outside it sounded just like an earthquake. Across from us there was actually a landslide—the debris flow was carrying huge boulders that crashed into the wall opposite our building. The landslide blocked the main river channel, and the water formed a barrier lake that in an instant submerged part of our rooms. We ran for our lives; a few people sprinted straight across to the slope on the other side. The ones who left a minute or two later found that the river water had already flooded our entire dormitory… the beds in our room were already floating, floodwater was pouring into the house. By the time we got out onto the lawn, the water was already enormous, and the current was so strong you couldn't stand. The only lucky thing was that the blockage of the main channel was broken open again by the torrent after about half an hour, and the water gradually drained away."

This event tested not only the reserve's infrastructure and emergency-response capability but also the ranger's psychological resilience. The reserve subsequently tightened its risk-averse procedures, strictly enforcing the rule "advance weather assessment during the rainy season, and absolutely no field work in extreme weather," and strengthening guidance for tourists who wade in the water in summer.

4.4 Value Identification: The Formation of Occupational Meaning in Ecological Stewardship

4.4.1 Family Support and the Sustained Motivation Behind the Career Choice

In the interview, the interviewee mentioned his family's supportive stance on his work. When asked "how does your family feel about your job," he said that "when I chose this line of work on my own, my family basically had no objection," explaining that the main reason was "being close to home, which makes it easier to take care of the family." This shows that family support plays an important part in a ranger's decision to enter the field and in his long-term persistence. For frontline rangers as a group, family support provides not only emotional encouragement but also a practical anchor that enables them to keep doing the work over the long run.

4.4.2 From Making a Living to Value Identification: "Doing What I Love"

The interview reveals that the interviewee's professional identification went through a developmental trajectory: from being driven by practical needs, to building an interest, and finally arriving at identification with the value of the work. His early choice to become a ranger stemmed

mainly from practical considerations such as being close to home and being able to take care of his family. Through long-term coexistence with the natural environment, he gradually developed an interest in wildlife and ecological protection, and eventually came to understand this career as a long-term choice of personal significance. This suggests that the professional identification of frontline ecological stewards does not come purely from external demands, but is continually formed in the interplay between life needs, work experience, and personal values.

5 Conclusion and Recommendations

Through an in-depth interview with a ranger at Tangjiahe Nature Reserve, this study portrays the work practice, occupational experience, and value identification of a frontline ecological steward along four dimensions—daily patrol, stories of stewardship, professional development, and value identification. The main conclusions are as follows. First, the work of frontline rangers is shifting from traditional patrol toward comprehensive ecological protection that encompasses ecological monitoring and disaster early warning. Second, the healing quality of long-term exposure to the natural environment comes alongside professional loneliness and an information gap with the wider world; natural imagery becomes a way of easing loneliness and regulating emotions in off-grid settings. Third, the ranger's identification with the value of his work moved from being driven by practical needs toward gradually recognizing the value of what he does.

Based on these conclusions, this paper proposes the following recommendations. First, build a sound system of psychological counseling and support for frontline rangers, providing the necessary organizational care for the off-grid conditions, loneliness, and disaster-related trauma they face in the field. Second, recognize rangers as "communicators of ecological culture" and involve them in public-facing activities such as nature education and ecological interpretation.

The daily work of a Tangjiahe ranger may seem ordinary, yet it forms an indispensable frontline practice in the operation of protected areas. These long-term, concrete acts of frontline stewardship are precisely the real-life embodiment of ecological-civilization construction in everyday life.

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